Beyond Smooth Spaces: Discrete Fisher Skeletons in Large Language Model Behavior

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Abstract

Large language models are usually described as moving through smooth, high-dimensional spaces: parameter space, activation space, representation manifolds, loss landscapes, and probability simplices. This paper argues that this smooth description is incomplete at the behavior level. When a model makes categorical predictions on labels with finite algebraic structure, such as weekdays or months, the output simplex after unembedding and Softmax becomes a place where a discrete arithmetic skeleton can be audited.

At maximum entropy, the Fisher tangent space splits into orthogonal packets indexed by finite additive characters and grouped by conductor. Away from maximum entropy, these packets no longer remain invisible to one another: their first structured coupling is governed by the raised Amari-Chentsov cubic tensor, and higher Taylor coefficients are organized by a finite quotient-character adjacency algebra. Natural gradient methods see the quadratic Fisher metric. The arithmetic skeleton begins where that quadratic approximation stops being faithful.

We test this claim on structured categorical behavior extracted from real transformer outputs. On the prime cyclic weekday task C_7, both Llama-3.1-8B and Mistral-7B-v0.3 preserve cyclic structure under Fisher/Hellinger-aware metrics while ordinary moment-Euclidean summaries collapse. On the composite month task $C_{12} \cong C_4 \times C_3$, Llama-3.1-8B shows a sharper moment collapse and a conductor-energy distribution close to the per-character-uniform null but tilted toward the high conductors 6 and 12. This is suggestive composite-control evidence, not yet a standalone proof of CRT-specific routing. An activation-pullback audit indicates that the weekday gap is not primarily an activation-space distortion. A corpus bigram control explains part of the observed directional asymmetry, but not the main metric gap.

This places the paper next to, rather than against, the mechanistic interpretability literature on grokking and modular arithmetic. Prior work has shown Fourier and representation-theoretic structure inside small networks trained on modular addition and finite group composition. The present paper asks the complementary behavior-side question: after a large pretrained transformer is projected through unembedding and Softmax onto a structured finite label set, does the output simplex still expose the corresponding finite Fourier/conductor structure as auditable behavior?

The practical thesis is modest and testable: structured categorical behavior can expose finite arithmetic control channels inside apparently smooth neural systems. These channels can be used as diagnostics for when Fisher or natural gradient approximations are overconfident, and as design constraints for update-side projection rules that suppress leakage outside declared conductor supports. The paper does not claim a universal theory of neural cognition, training dynamics, or intelligence. It establishes a sharper behavior-level object: discrete Fisher skeletons in large language model outputs.

Keywords: large language models, information geometry, Fisher geometry, Hellinger geometry, natural gradient, Amari-Chentsov tensor, categorical simplex, conductor packets, finite Fourier analysis, quotient characters, AI optimization, behavior manifolds.

Contents

1. The Claim . . . 2
2. Why Smooth Geometry Is Not Enough . . . 3
3. The Finite Skeleton . . . 4
4. The Cubic Correction Natural Gradient Misses . . . 5
5. Why Language Models Should Expose the Skeleton . . . 6
6. Relation to Grokking and Modular Arithmetic . . . 6
7. Empirical Tests . . . 7
    7.1 Weekdays: the prime C_7 control . . . 7
    7.2 Months: the composite C_{12} control . . . 8
    7.3 Activation pullback audit . . . 9
    7.4 Bigram confound control . . . 10
    7.5 Optimizer pilots . . . 10
8. What the Evidence Shows . . . 11
9. What the Evidence Does Not Show . . . 11
10. Consequences for AI Diagnostics . . . 12
11. Conclusion . . . 12

Data and Code Availability . . . 13
Acknowledgments . . . 13
References . . . 13
Appendix A: Notation and Minimal Definitions . . . 14
    A.1 The simplex and tangent space . . . 14
    A.2 Fisher metric . . . 14
    A.3 Characters and conductors . . . 15
    A.4 Hellinger coordinates . . . 15
    A.5 Raised Amari-Chentsov cubic . . . 15
Appendix B: Empirical Artifact Ledger . . . 15
    B.1 Weekday behavior reports . . . 16
    B.2 Month behavior reports . . . 16
    B.3 Activation pullback reports . . . 16
    B.4 Corpus confound reports . . . 16
    B.5 Optimizer and projection reports . . . 16
Appendix C: Falsifiers and Next Measurements . . . 17
Appendix D: Scope Boundaries . . . 17
Appendix E: Pandoc and LaTeX Notes . . . 18

1. The Claim

The usual geometric language for modern AI is smooth. We speak about continuous parameter spaces, differentiable loss landscapes, activation manifolds, representation geometry, natural gradients, and local quadratic approximations. That language is necessary. It is not sufficient.

The reason is simple: the behavior of a language model is categorical. A model returns a probability distribution over tokens, answers, labels, bins, or concepts. If those labels carry structure, then the probability simplex is not merely a featureless smooth manifold. It is also a finite statistical object with arithmetic coordinates.

This paper chooses a specific locus of investigation: the output simplex Δ°_R after unembedding

and Softmax. That choice is deliberately weaker than claiming that the skeleton primarily lives there. Pre-Softmax logits or internal activations may carry a sharper version of the same finite structure. The point of the output simplex is that it is where the structure becomes auditable as categorical behavior: a probability distribution with a Fisher metric, a Hellinger embedding, and finite character coordinates tied to the label algebra.

This paper studies the following claim.

Claim. Let a model produce probability distributions on a finite structured label set R. If R has finite algebraic structure, then the behavior simplex Δ_R° carries a discrete Fisher skeleton:

1. at maximum entropy, Fisher tangent directions split into orthogonal quotient-character packets;
2. those packets are grouped by conductor, giving a finite arithmetic resolution of behavior;
3. away from maximum entropy, the first structured correction to the quadratic Fisher picture is the raised Amari-Chentsov cubic;
4. real transformer outputs on structured tasks expose this skeleton in measurable metric gaps and conductor-energy decompositions.

The precise meaning is not that primes explain all cognition. It is that finite categorical behavior can decompose into prime and prime-power channels before it becomes a smooth continuum. In such settings, arithmetic is not decoration on top of a neural manifold. It is part of the behavior geometry.

2. Why Smooth Geometry Is Not Enough

Let $p \in \Delta_R^\circ$ be a probability distribution over a finite set R, with

$$\Delta_R^\circ = \left\{ p \in \mathbb{R}_{>0}^R : \sum_{a \in R} p_a = 1 \right\}.$$

The tangent space at any point is the zero-sum hyperplane

$$T_p \Delta_R^\circ = H_R = \left\{ u \in \mathbb{R}^R : \sum_{a \in R} u_a = 0 \right\}.$$

The Fisher metric is

$$g_p(u, v) = \sum_{a \in R} \frac{u_a v_a}{p_a}.$$

At the maximum-entropy point $p_* = |R|^{-1}\mathbf{1}$, this becomes

$$g_{p_*}(u, v) = |R| \sum_{a \in R} u_a v_a.$$

Thus the local geometry at p_* is a scaled Euclidean geometry on the zero-sum hyperplane. This is the quadratic geometry used by Fisher preconditioning and natural gradient methods. It tells us how expensive an infinitesimal displacement is.

But the Fisher metric is only the second-order object. A categorical exponential family also has a third-order tensor, the Amari-Chentsov tensor. In raised form we write

$$K^{CH} = \Gamma^{(m)} - \Gamma^{(e)},$$

where $\Gamma^{(m)}$ and $\Gamma^{(e)}$ are the mixture and exponential connection coefficients. This cubic tensor measures the first failure of the two flat affine structures to agree. Operationally, it detects when a straight line in one coordinate system is no longer locally faithful to the Fisher/Hellinger geometry of the probability simplex.

Natural gradient sees g. The behavior skeleton begins to move at K^{CH}.

This distinction matters for AI because a model trained by cross-entropy does not remain near maximum entropy. It becomes useful by leaving the centroid. The same movement that makes a model informative also activates cubic behavior-side geometry.

3. The Finite Skeleton

Assume now that R is not just a finite set, but a finite abelian group or finite quotient ring. The examples in this paper are deliberately small:

$$C_7 = \mathbb{Z}/7\mathbb{Z}$$

for weekdays, and

$$C_{12} = \mathbb{Z}/12\mathbb{Z} \cong C_4 \times C_3$$

for months.

The additive characters of R provide a finite Fourier basis. For the cyclic case C_N, a character is

$$\chi_k(a) = \exp(2\pi i k a/N), \qquad k \in C_N.$$

The trivial character $k = 0$ spans the normalization direction. The nontrivial characters span the Fisher tangent space at p_*. Their conductors group the tangent directions into arithmetic packets. In the cyclic case the conductor of k is

$$\mathrm{cond}(k) = \frac{N}{\gcd(k, N)}.$$

At p_*, these packets are Fisher-orthogonal. This gives a finite decomposition

$$H_R \otimes \mathbb{C} = \bigoplus_{\mathfrak{f} \neq 1} X_{\mathfrak{f}},$$

where $X_{\mathfrak{f}}$ is the conductor packet associated with conductor $\mathfrak{f}$.

For C_7, there is only one nontrivial conductor packet: conductor 7. This makes weekdays the prime control. There is no CRT split and no mixed prime structure.

For C_{12}, there are multiple conductor packets:

$$2, \quad 3, \quad 4, \quad 6, \quad 12.$$

This makes months the composite control. The Chinese Remainder Theorem predicts that behavior can distribute energy across separate 2-primary, 3-primary, and mixed conductor channels.

The smooth simplex has not disappeared. It is still the right ambient statistical manifold. The point is that structured labels add a finite arithmetic atlas on its tangent and higher-order geometry.

This is also where the companion simplicial algebra paper matters. That paper works on the zero-sum hyperplane as a native coordinate object: an overcomplete simplicial frame modulo the gauge direction $\mathbf{1}$. The probability simplex has the same linear tangent arena. At p_*, the Fisher metric is just a scale multiple of the simplicial zero-sum metric:

$$g_{p_*}(u, v) = |R| \sum_{a \in R} u_a v_a.$$

Thus the companion paper supplies the algebraic gauge language for the space on which the present paper measures behavior. In the $N = 4$ row, its cross-product and projector identities also give a concrete finite model for how a symmetric cubic can live intrinsically on a zero-sum simplex slice. The present paper does not need that $N = 4$ exception for the weekday and month experiments, but it uses the same organizing principle: the meaningful geometry is not the ambient coordinate box; it is the zero-sum statistical hyperplane with its native finite symmetries.

4. The Cubic Correction Natural Gradient Misses

At maximum entropy, the Fisher metric diagonalizes cleanly in the character basis. Off maximum entropy, the packets begin to couple. For a displacement

$$h = p - p_*,$$

the categorical cubic response at the centroid has the convolution form

$$K_{p_*}^{CH}(h)v = |R|\, h * v - |R|\, \langle h, v \rangle p_*,$$

where $*$ denotes convolution in the finite output algebra, and the final term removes the normalization direction.

This formula is the first place the finite algebra becomes dynamical. In Fourier coordinates, convolution becomes multiplication of characters. The cubic tensor therefore moves energy between character packets according to finite arithmetic rules. For prime cyclic C_7, all nontrivial characters belong to the same conductor packet. For composite C_{12}, products of characters can cross between conductor packets in CRT-structured ways.

The conclusion is not that natural gradient is wrong. Natural gradient is the correct quadratic local approximation. The conclusion is that the quadratic approximation is blind to structured cubic response. When the model moves far enough from maximum entropy, a Fisher-aware optimizer needs a warning light: is the local quadratic metric still faithful, or is the cubic response already large and structured?

This is the role of K^{CH} in the present paper. It is a behavior-side curvature diagnostic.

5. Why Language Models Should Expose the Skeleton

For a labeled example with target y, cross-entropy has logit gradient

$$\nabla_z L = p - e_y.$$

Near p_*, this is

$$p - e_y = h - (e_y - p_*),$$

where $h = p - p_*$. Thus training repeatedly injects centered score-frame directions

$$e_y - p_* \in H_R.$$

If the labels carry finite algebraic structure, those score-frame directions already have a quotient-character decomposition. The model's continuous parameter dynamics decide which packets are easy to move, but the behavior side supplies the finite coordinate system.

This gives the main AI-facing hypothesis.

Hypothesis. On structured categorical tasks, a trained transformer's output distributions should preserve Fisher/Hellinger geometry more faithfully than ordinary moment-Euclidean summaries, and composite tasks should reveal whether displacement energy is merely dimension-weighted across characters or is additionally tilted by the finite conductor structure.

This is a behavior-level hypothesis. It does not require that every hidden unit or training trajectory explicitly factor into conductors. It only claims that when the model's outputs are projected onto a structured finite label set, the resulting probability distributions can expose the arithmetic skeleton of the output simplex.

6. Relation to Grokking and Modular Arithmetic

The closest existing dialogue partner is the mechanistic interpretability literature on grokking. Power et al. introduced grokking as delayed generalization beyond overfitting on small algorithmic tasks [Power-Burda-Edwards-Babuschkin-Misra]. Nanda et al. then reverse engineered small transformers trained on modular addition, showing that the learned algorithm uses discrete Fourier transforms and trigonometric identities, and validating the circuit with Fourier-space ablations [Nanda-Chan-Lieberum-Smith-Steinhardt]. Chughtai, Chan, and Nanda extended this line to finite group composition, showing that small networks can implement group operations using representation-theoretic structure [Chughtai-Chan-Nanda].

Those papers look inside trained networks. This paper looks at the behavior simplex after a model has already produced categorical probabilities. The two views are complementary:

1. mechanistic work asks whether weights and activations implement Fourier or group-representation algorithms;
2. the present behavior-side test asks whether the final output distributions retain conductor-graded Fisher geometry.

The strongest future synthesis would measure both at once: decompose internal Fourier features at a chosen layer, decompose output displacements into conductor packets, and test whether internal packet energy predicts output-side conductor energy after the unembedding and Softmax channel.

This distinction also clarifies a possible skeptical reading of the month experiment. If logit-space distances track cyclic structure better than Fisher/Hellinger distances after Softmax, that does not refute the behavior claim. It suggests that some conductor structure may be sharper before the final measurement channel and partially compressed by Softmax. In that case the output simplex is not the only place the structure lives; it is the place where the structure becomes auditable as categorical behavior. Closing this loop requires the paired internal/output experiment above.

7. Empirical Tests

We use two structured output tasks.

The first is the weekday task, modeled as C_7. It is the prime control: there is one nontrivial conductor packet, so the test is mainly a metric test. Do Fisher/Hellinger-aware distances preserve the cyclic behavior geometry when moment-Euclidean summaries collapse?

The second is the month task, modeled as C_{12}. It is the CRT control: there are multiple conductor packets, so the test is both a metric test and an energy-decomposition test. Does the empirical displacement energy distribute across the predicted conductors?

All measurements reported here are centroid-level behavior measurements. They are not yet a full training-time audit of how conductor packets emerge during optimization.

7.1 Weekdays: the prime C_7 control

For weekdays, the output algebra is C_7. We measured per-class output distributions for Llama-3.1-8B and Mistral-7B-v0.3 at layer 28, last token, and compared pairwise behavior distances to the cyclic arc distance.

The key correlation table is:

Metric	Llama-3.1-8B	Mistral-7B-v0.3	symmetric proxy
Hellinger chord vs cyclic arc	0.899815	0.916873	0.910010
moment-Euclidean chord vs cyclic arc	0.614547	0.714715	0.886448
natural/logit chord vs cyclic arc	0.944337	0.955300	0.941295

Metric	Llama-3.1-8B	Mistral-7B-v0.3	symmetric proxy
Fisher Levi-Civita distance vs cyclic arc	0.903176	0.922202	0.911776

The pattern is stable across two model families. Moment-Euclidean summaries pay the largest price. Fisher/Hellinger-aware metrics preserve the cyclic structure.

This is exactly what the Fisher-skeleton hypothesis predicts in the prime case. Since C_7 has only one nontrivial conductor packet, the relevant failure mode is not a split across multiple conductors. It is the failure of linear moment geometry to represent the curved behavior simplex.

7.2 Months: the composite C_{12} control

For months, the output algebra is

$$C_{12} \cong C_4 \times C_3.$$

This task contains multiple conductors. The measured Llama-3.1-8B correlations are:

Metric	real Llama-3.1-8B	symmetric proxy
Hellinger chord vs cyclic arc	0.708696	n/a
moment-Euclidean chord vs cyclic arc	0.210147	0.696128
natural/logit chord vs cyclic arc	0.870898	n/a
Fisher Levi-Civita distance vs cyclic arc	0.713899	n/a

The moment summary collapses much more sharply than in the weekday case. The same measurement also gives a conductor-energy decomposition. Averaged over the 12 input classes, the empirical displacement energy is close to the per-character-uniform null determined by packet dimensions $(1, 2, 2, 2, 4)$, but with a small systematic high-conductor tilt:

conductor	char. count	dimension null	observed	observed - null
2	1	0.0909	0.0856	−0.0053
3	2	0.1818	0.1689	−0.0129
4	2	0.1818	0.1730	−0.0088
6	2	0.1818	0.1921	+0.0103
12	4	0.3636	0.3804	+0.0168

This table changes the interpretation. The aggregate split should not be described, by itself, as proof that the model learned CRT-specific routing. Most of the distribution is explained by packet dimension: conductor 12 has four characters, conductors $3, 4, 6$ have two, and conductor 2 has one. What remains is a modest but stable tilt away from conductors $2, 3, 4$ and toward $6, 12$.

A row bootstrap over the 12 input classes gives 95% confidence intervals for the differences from the dimension null:

conductor	bootstrap 95% CI for observed - null
2	$[-0.0102, -0.0013]$
3	$[-0.0199, -0.0071]$
4	$[-0.0155, -0.0036]$
6	$[+0.0035, +0.0190]$
12	$[+0.0088, +0.0260]$

A common output-label permutation null, which preserves the class probability spectra while breaking the alignment with the C_{12} character basis, places the observed conductor 12 fraction at the 98.7th percentile and the aggregate L^2 distance from the dimension null at the 98.0th percentile. This supports a cautious composite-control reading: the measured month behavior is not random with respect to the C_{12} basis, but the aggregate conductor table alone is not enough to claim CRT routing.

The class-level mixed conductor 6 row is more diagnostic. Its mean is 0.1921 against the dimension null $2/11 = 0.1818$, but the largest class-level elevations come from January (+0.045) and April (+0.035). Thus the next empirical question is class anisotropy, not just aggregate energy. Prompt-level bootstraps, an unstructured 12-class control, and a second model family are required before the composite result can carry the same weight as the C_7 replication.

The natural/logit chord result is also important. On months it correlates with the cyclic arc more strongly than Hellinger or Fisher endpoint distance. This is consistent with the view that pre-Softmax coordinates retain a sharper cyclic signal, while Softmax maps that signal into the curved probability simplex where Fisher/Hellinger geometry becomes the correct output-side metric. Thus the logit result is not a nuisance to hide; it is evidence that the conductor structure may straddle both sides of the final measurement channel. A paired internal-output conductor audit is the right test.

7.3 Activation pullback audit

One possible objection is that the metric gap might come from activation space rather than from the output simplex. Let

$$F : A \to \Delta_R^\circ$$

be the learned activation-to-behavior map. The quadratic pullback metric is

$$J_F^\top g_y J_F.$$

If this pullback varies strongly across class pairs, then the metric gap could be dominated by activation-side distortion rather than behavior-side Fisher geometry.

The Llama-3.1-8B activation pullback audit gives:

task	activation distance vs Hellinger	activation distance vs cyclic arc	pullback coefficient of variation
weekdays	0.936878	0.900147	0.1540
months	0.821285	0.651696	0.2331

For weekdays, the activation centroids are close to a rescaled isometry of the Fisher/Hellinger output geometry. This supports the interpretation that the weekday metric gap is primarily intrinsic to the behavior simplex. For months, the activation map still preserves Hellinger geometry substantially, but the larger coefficient of variation indicates that activation-side distortion contributes more visibly.

This is a single-layer audit at layer 28. It should not be interpreted as a layer-invariant theorem. A layer sweep is a required robustness check before submission, especially because different layers may carry different mixtures of internal Fourier structure, task abstraction, and final readout geometry.

7.4 Bigram confound control

The weekday and month outputs also show directional asymmetries. For example, some weekday classes leak more probability toward their cyclic predecessor than toward their successor. Such asymmetry could reflect arithmetic off-centroid structure, but it could also reflect ordinary language exposure: "Sunday Monday", "Monday Tuesday", and related bigram patterns.

To test this, a C4 corpus bigram control was run on 10 million whitespace tokens. The registered weekday criterion for "fully explain" required both large correlation and high sign agreement. The control landed in the intermediate band:

target	best qualitative verdict
Llama weekdays	intermediate
Mistral weekdays	intermediate
Llama months	intermediate

The consequence is important. Directional predecessor leakage should not be used as independent proof of arithmetic carry. It is consistent with a mixture of learned language priors and residual cyclic structure.

The main support for the present paper is therefore not the asymmetry alone. It is the metric gap, the prime/composite contrast, the C_{12} null-controlled high-conductor tilt, and the activation-pullback audit.

7.5 Optimizer pilots

The conductor picture also suggests update-side controls. If a task is supposed to remain inside a closed conductor support, then an optimizer can project the applied update into that support while leaving the stochastic gradient and moments unprojected. The clean rule is:

project the applied update, not the raw gradient.

In a synthetic $\mathbb{F}_4[u]/(u^2)$ conductor-closed optimizer pilot, full update projection improved the measured KL relative to the unprojected setting, from 0.036156 to 0.023031. The corresponding machine-precision leakage value is only a sanity check that the projection operator projected into its declared support; it is not independent evidence for the hypothesis.

In the calendar-transformer month pilots, update-only conductor Adam also enforced machine-precision leakage relative to declared supports. The most relevant run is the width-64 offset-OOD split with mixed support $\{6, 12\}$: training examples use offsets $1, \ldots, 6$, while test examples use unseen offsets $7, \ldots, 12$. Against Adam, update-only conductor Adam improved test accuracy by $+0.013889$, reduced the generalization gap by 0.203329, and reduced test loss from 8.694958 to 8.570058, while leaving raw gradients and Adam moments unprojected. By contrast, the conductor-12-only offset-OOD run hurt test accuracy by -0.027778, and the full-support run is the identity control.

These are small pilot numbers, but they are not merely "the projection projected." They suggest that the choice of conductor support can matter on a held-out structured split. The evidence should still be read as preliminary: strong support for leakage control, suggestive support for support selection, not yet a general-purpose optimizer theorem.

8. What the Evidence Shows

The current evidence supports four statements.

First, structured categorical outputs can expose Fisher/Hellinger geometry that ordinary moment-Euclidean summaries miss. This is replicated on the weekday C_7 task across Llama-3.1-8B and Mistral-7B-v0.3.

Second, composite structured outputs can be audited against conductor structure. The month C_{12} task is close to the per-character dimension null but shows a small stable tilt toward conductors 6 and 12. This is suggestive composite-control evidence, not yet proof of CRT-specific routing.

Third, the weekday metric gap is not primarily explained by activation distortion at the audited layer. The activation pullback is close to a rescaled isometry of the output Hellinger geometry on the class-centroid set.

Fourth, update-side conductor projection can suppress declared leakage to machine precision in pilots. This is already useful as a diagnostic and as a design pattern, even before it becomes a mature optimizer.

9. What the Evidence Does Not Show

The evidence does not show that all neural cognition is arithmetic.

It does not show that stochastic gradient descent universally decomposes into conductor packets during training. The current measurements are mostly final behavior probes and small optimizer pilots. A full training-time audit remains future work.

It does not show that K^{CH} detects every filtered obstruction. The cubic tensor is a visible Fisher shadow. Some extension data requires the operator-valued quotient-character adjacency algebra rather than scalar conductor summaries.

It does not show that linguistic confounds are absent. The bigram control shows that sequence priors explain part of the directional asymmetry. The stronger evidence is metric and null-controlled

conductor-energetic, not merely directional.

It does not show that a conductor-projected optimizer is ready to replace Adam. The optimizer evidence shows leakage control and promising local rows. It is not yet a broad training benchmark.

It also does not yet separate two possible sources of the measured structure. One possibility is behavior-simplex universality: any accurate model on a structured categorical task will expose the same conductor geometry because the output simplex and label algebra force it. Another possibility is learned representation specificity: only models that internally discover matching Fourier or representation-theoretic features will show the conductor split cleanly. The decisive experiment is paired: measure internal Fourier/conductor energy at several layers and output-side conductor energy on the same prompts, then compare against label-permuted and unstructured 12-class controls.

These boundaries are not cosmetic. They are the reason the narrower claim is strong:

> Some large language model behavior manifolds contain measurable discrete Fisher skeletons.

That sentence is already enough.

10. Consequences for AI Diagnostics

The practical diagnostic question is:

> When is the quadratic Fisher approximation locally faithful, and when is it flattening structured cubic response?

This can be turned into concrete measurements.

1. Choose a structured output algebra R.
2. Measure per-class output distributions.
3. Compare moment-Euclidean, logit, Hellinger, and Fisher distances.
4. Decompose displacements $h_i = p_i - p_*$ into conductor packets.
5. Estimate $K^{CH}_{p_*}(h_i)$ and cubic step fractions.
6. Audit activation pullback distortion via $J_F^\top g_y J_F$.
7. Run corpus or prompt controls for obvious confounds.

If the moment geometry collapses while Fisher/Hellinger geometry remains stable, and if composite tasks distribute energy across predicted conductors, then the behavior manifold is not being faithfully described by a featureless smooth picture.

This matters for interpretability because it supplies a finite set of behavior-side channels. It matters for optimization because it supplies a finite warning system for overconfident quadratic steps. It matters for model auditing because it separates three things that are often conflated:

1. output-side information geometry;
2. activation-side pullback distortion;
3. linguistic or dataset confounds.

11. Conclusion

Large language models are continuous systems trained by continuous optimization, but their behavior is often categorical. When those categories carry finite algebraic structure, the behavior simplex

after Softmax can expose more than a smooth Fisher manifold. It can expose a finite arithmetic skeleton.

At maximum entropy, this skeleton appears as Fisher-orthogonal quotient-character conductor packets. Away from maximum entropy, the Amari-Chentsov cubic gives the first structured correction that natural gradient misses. In real transformer outputs, the skeleton is visible most cleanly in the prime control: weekday behavior preserves Fisher/Hellinger geometry while moment summaries collapse across two model families. The composite month control adds a sharper moment collapse and a small, stable high-conductor tilt relative to the per-character dimension null; it is a promising but not yet decisive CRT-routing signal.

The conclusion is deliberately scoped. This is not a universal claim about all intelligence. It is a sharper claim about structured categorical behavior:

> Beyond smooth spaces, large language model behavior can contain discrete Fisher skeletons.

That is already a new object for AI diagnostics, optimization, and interpretability.

Data and Code Availability

The empirical artifacts summarized in this paper are stored in the companion GitHub repository for this project. The key checked-in reports are listed in Appendix B. The repository contains the scripts, report markdown files, JSON payloads, and local artifact paths used to produce the tables above. Some large model-output tensors are stored as local safetensor artifacts and should be mirrored or regenerated before archival release.

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Appendix A: Notation and Minimal Definitions

A.1 The simplex and tangent space

For a finite output set R, the open probability simplex is

$$\Delta_R^\circ = \left\{ p \in \mathbb{R}_{>0}^R : \sum_{a \in R} p_a = 1 \right\}.$$

The tangent space is the zero-sum hyperplane

$$H_R = \left\{ u \in \mathbb{R}^R : \sum_{a \in R} u_a = 0 \right\}.$$

The maximum-entropy point is

$$p_* = |R|^{-1} \mathbf{1}.$$

A.2 Fisher metric

For $u, v \in H_R$, the Fisher metric is

$$g_p(u, v) = \sum_{a \in R} \frac{u_a v_a}{p_a}.$$

At p_*,

$$g_{p_*}(u, v) = |R| \sum_{a \in R} u_a v_a.$$

This is the point at which the zero-sum simplicial gauge language of the companion algebra paper becomes identical, up to scale, with the Fisher tangent geometry of the categorical simplex.

A.3 Characters and conductors

For $C_N = \mathbb{Z}/N\mathbb{Z}$, the additive character indexed by k is

$$\chi_k(a) = \exp(2\pi i k a/N).$$

The trivial character $k = 0$ corresponds to the normalization direction. The nontrivial characters span $H_R \otimes \mathbb{C}$. The conductor of χ_k is

$$\mathrm{cond}(k) = \frac{N}{\gcd(k, N)}.$$

Characters with the same conductor form a conductor packet.

A.4 Hellinger coordinates

The square-root map

$$p \mapsto \sqrt{p}$$

embeds the open simplex into the positive orthant of the unit sphere. This is why Hellinger distances are Fisher-aware rather than arbitrary Euclidean features. The Hellinger chord distance is the Euclidean chord distance after the square-root embedding, up to the conventional normalization scale.

A.5 Raised Amari-Chentsov cubic

The raised Amari-Chentsov tensor is written

$$K^{CH} = \Gamma^{(m)} - \Gamma^{(e)}.$$

At the centroid of a finite group algebra, its first convolution form is

$$K^{CH}_{p_*}(h)v = |R|\, h * v - |R|\, \langle h, v\rangle p_*.$$

This is the cubic behavior-side correction emphasized in the main text.

Appendix B: Empirical Artifact Ledger

This appendix lists the local reports behind the main empirical tables. The paths are repository-relative. All files listed below live under `empirical/reports/`; filenames are given without repeating that directory in every row so the PDF tables stay readable.

B.1 Weekday behavior reports

- Llama-3.1-8B C_7 metric gap: `weekdays_K_CH_real_llama31_8b.md`.
- Mistral-7B-v0.3 C_7 metric gap: `weekdays_K_CH_real_mistral_7b.md`.

The key result is cross-family replication: Fisher/Hellinger-aware metrics remain near the cyclic geometry while moment-Euclidean summaries fall.

B.2 Month behavior reports

- Llama-3.1-8B C_{12} metric gap: `months_K_CH_real_llama31_8b.md`.
- C_{12} conductor energy fractions: `months_K_CH_real_llama31_8b.md`.
- C_{12} per-character and permutation null comparison: the `months_conductor_null` report in `empirical/reports`.

The raw aggregate conductor split is:

$$(0.0856,\ 0.1689,\ 0.1730,\ 0.1921,\ 0.3804)$$

for conductors $2, 3, 4, 6, 12$, respectively. The null report compares this against the packet-dimension null $(0.0909, 0.1818, 0.1818, 0.1818, 0.3636)$ and records the small high-conductor tilt toward 6 and 12.

B.3 Activation pullback reports

- Weekday activation pullback: `activation_pullback_weekdays_llama31_8b.md`.
- Month activation pullback: `activation_pullback_months_llama31_8b.md`.

The weekday pullback coefficient of variation is 0.1540, supporting the interpretation that the weekday metric gap is primarily behavior-side.

B.4 Corpus confound reports

- C4 10M bigram summary: `bigram_corpus_control_c4_10M_summary.md`.
- Llama weekday detail: `bigram_corpus_control_weekdays_llama31_8b_c4_10M.md`.
- Mistral weekday detail: `bigram_corpus_control_weekdays_mistral_7b_c4_10M.md`.
- Llama month detail: `bigram_corpus_control_months_llama31_8b_c4_10M.md`.

The control lands in the intermediate band. It partially explains directional asymmetry and prevents overclaiming from predecessor leakage alone.

B.5 Optimizer and projection reports

- Synthetic conductor-closed optimizer: `conductor_closed_optimizer_f4_u2.md`.
- Projection Pareto sweep: `conductor_projection_pareto_f4_u2.md`.
- Month calendar transformer default split: `conductor_calendar_transformer_months.md`.
- Month calendar transformer offset-OOD, mixed support: `conductor_calendar_transformer_months_d64_offset_ood_c6_c12.md`.

The mixed-support offset-OOD pilot improves test accuracy by +0.013889 and reduces the generalization gap by 0.203329 against Adam on the same split. The optimizer evidence supports

leakage control and makes support selection worth testing. Broader performance claims require larger benchmarks.

Appendix C: Falsifiers and Next Measurements

The discrete Fisher skeleton hypothesis is meant to be falsifiable. The following outcomes would weaken or narrow it.

1. On replicated structured tasks, Fisher/Hellinger distances collapse at the same rate as moment-Euclidean summaries.
2. Month-like composite tasks reduce entirely to the per-character-uniform null, with no stable high-conductor tilt and no class-level anisotropy.
3. Label-permutation and unstructured 12-class controls reproduce the same C_{12} conductor pattern.
4. Month-like composite tasks fail to show any stable conductor-energy decomposition across model families or prompt sets.
5. Activation-pullback distortion explains essentially all of the metric gap.
6. Corpus controls fully explain the observed geometry, including the metric gap and conductor-energy pattern, not merely directional asymmetry.
7. Training-time checkpoint audits show no conductor emergence from initialization through convergence.
8. Update-side conductor projections suppress leakage only by destroying task performance under fair baselines.

The next high-value measurements are:

1. replicate the C_{12} month null-controlled high-conductor tilt on a second model family;
2. add prompt-level bootstrap confidence intervals for the C_{12} conductor fractions;
3. run an unstructured 12-class control, such as colors or foods;
4. run layer sweeps for both weekdays and months;
5. run a paired internal-output conductor audit comparing hidden Fourier features with output-side conductor energy;
6. run checkpoint audits from initialization to convergence on a small transformer trained on structured labels;
7. mirror or regenerate all large safetensor artifacts for public archival release.

Appendix D: Scope Boundaries

This paper uses the following restricted meanings.

Behavior manifold. The finite set of measured output distributions and the Fisher/Hellinger geometry they inherit from the probability simplex.

Discrete Fisher skeleton. The character/conductor decomposition of the Fisher tangent space at maximum entropy, together with its off-centroid cubic and finite-adjacency corrections.

This paper assumes the algebraic frame of [Murillo-Simplicial] and tests its statistical instantiation on model behavior. The fuller dually-flat (θ, η) development of simplicial coordinates is treated

elsewhere; it is not the main claim of this paper.

Conductor packet. A group of nontrivial additive characters sharing the same conductor.

Cubic diagnostic. The use of K^{CH} to estimate when the quadratic Fisher approximation is missing structured behavior-side response.

Not claimed. The paper does not claim that every neural representation is arithmetic, that every task has a useful finite quotient structure, that ordinary Adam should be replaced, or that final-output measurements by themselves prove training-time conductor dynamics.

Appendix E: Pandoc and LaTeX Notes

This manuscript is written as Pandoc-compatible Markdown with LaTeX math. Displayed equations use

```
\[
...
\]
```

and inline equations use `\(...\)`. Tables are plain Markdown pipe tables. The top title block follows the same convention as the companion manuscript: an H1 title, a bold author line, an italic date line, an abstract block, and a horizontal rule before the main body.